

Apr 16, 2026

Human CSF Processing for 10x scRNA-seq

DOI

<https://dx.doi.org/10.17504/protocols.io.eq2lyn63rvx9/v1>

Le Zhang¹

¹Yale University

ASAP Collaborative Res...



Nicholas C Buitrago-Pocasangre

Yale University

OPEN  ACCESS



DOI: <https://dx.doi.org/10.17504/protocols.io.eq2lyn63rvx9/v1>

Protocol Citation: Le Zhang 2026. Human CSF Processing for 10x scRNA-seq. **protocols.io**

<https://dx.doi.org/10.17504/protocols.io.eq2lyn63rvx9/v1>

License: This is an open access protocol distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working

We use this protocol and it's working

Created: March 28, 2022

Last Modified: April 16, 2026

Protocol Integer ID: 59971

Keywords: 10x CSF Single Cell, CSF Cells Freezing, Cerebrospinal Fluid (CSF), ASAPCRN, team hafler csf processing protocol for 10x, team hafler csf processing protocol, single cell rna, human csf processing

Funders Acknowledgements:

ASAP

Grant ID: ASAP-000529



Abstract

This protocol details about team Hafler CSF processing protocol for 10x single cell RNA sequencing.

Attachments



[392-855.docx](#)

17KB

Materials

Materials

- 10x Genomics Chromium Single Cell 5' Reagent Kit v2 chemistry
- Bambanker cell freezing medium
- Sprotte atraumatic lumbar needle with cannula
- CSF WASTE tube
- Swing bucket centrifuge
- C-Chip disposable hemacytometer (DHCN015)
- Mr. Frosty freezing container



CSF Collection

- 1 Collect the cerebrospinal Fluid (CSF) Room temperature (64-77°F / 18-25°C). No fasting is required. Once collected, keep samples at On ice or at 4 °C .

- 2 Perform lumbar puncture (LP) utilizing Sprotte atraumatic lumbar needle with cannula.

Note

In case of bleeding at the puncture site, collect the first 1-2 mL of CSF fluid in the CSF WASTE tube.

- 3 Continue collecting CSF fluid in the CSF WASTE tube until CSF is visibly devoid of red blood cells (RBC) contamination, i.e., no pinkish or red color. For the CSF sample, collect 30 mL of CSF fluid into pre-labeled tubes. Proceed with the next steps even if less than 30 mL of CSF was collected. Replace caps after CSF sample collection and mix gently by inverting each tube 3-4 times. CSF WASTE tube may be discarded.
- 4 Record the total CSF volume and if CSF sample looks grossly or visibly contaminated with RBC and/or if it is hued pink or red. Record the date and time of the CSF sample collected.
- 5 Transfer the CSF sample as quickly as possible to the lab. Then proceed to CSF centrifugation and processing.



10x CSF Single Cell Protocol

- 6 Spin down the CSF cells at 300 x g, 4°C, 00:10:00 using a swing bucket centrifuge.

10m



- 7 Aspirate the supernatant with a pipette, leaving ~ 500 µL without disturbing the cell pellet. Save the CSF supernatant On ice for further aliquots.

- 8 Mix and count the total CSF cell number. Use 5 µL of the sample with 1:1 dilution of trypan blue and a C-Chip disposable hemacytometer (DHCN015). Record the total CSF cell number.





9 Transfer to a 1.5-mL Eppendorf tube and spin down at  300 x g, 4°C, 00:10:00 using a swing bucket centrifuge.

10m



10 Carefully aspirate the supernatant, leaving ~  50 µL .

11 Resuspend the cells and count.

11.1 If the cells need to be diluted more, use the CSF supernatant.

11.2 If there are leftover CSF cells after 10x loading, freeze down the cells.

12 Submit for 10x single cell loading. Ideally 700-1200 cells/ul. If too few cells, submit all the sample of  50 µL .

13 Use 10x Genomics Chromium Single Cell 5' Reagent Kit v2 chemistry for single cell gene expression and TCR/BCR sequencing.

CSF Cells Freezing Protocol

5m

14 Spin down the CSF cells at  400 x g, 4°C, 00:05:00 using a swing bucket centrifuge.

5m



15 Carefully aspirate the supernatant.

16 Resuspend the cell pellet in  0.1 mL of Bambanker cell freezing medium.

17 Freeze in Mr. Frosty freezing container in  -80 °C freezer. Transfer to liquid N2 tank after 24h - 72h.

18 Transfer to liquid N2 tank after 24h - 72h.

